VTPEH 6109 - study of hypertensive disease in the USA

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1 Objective and Approach

1.1 Specific Objective

We will look at the impact of hypertension in men and women as they age. The CDC data gives the rate of hypertensive disease (as opposed to actual counts) in adults 20 and older, per gender, poverty level, age group, race. We will focus on exploring hypertension disease in relation to gender, age, and poverty level.
* Does hypertension affect men and women in the same fashion?
* Does hypertensive disease affect all individuals the same regardless of income?

https://data.cdc.gov/National-Center-for-Health-Statistics/DQS-Hypertension-in-adults-age-20-and-older-by-sel/va5e-efw9/about_data

```
#set working directory and read data file

data <-
  read.csv("../data/Hypertension_in_adults_age_20_and_older.csv", header = TRUE)
```

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(tibble)
```

1.2 Hypertension as function of age and gender

Does hypertension affect men and women in the same fashion?

1.2.1 Parsing

The focus is on data that illustrates the impact of hypertensive disease by age group, gender and for two time periods, 1988-1994 & 2017-3/2020

```
# simplify the data frame to keep only the data of interest
data_poverty <- subset(data, GROUP == "Poverty level" & ESTIMATE_TYPE ==
                        "Percent of population, age adjusted")
data_age <- subset(data, GROUP == "Sex and age group")

# remove redundant data
data_age <- subset(data_age, !(SUBGROUP_ORDER %in% c(13,16,21,24)) )
# remove columns of no use for this study
data_age <- data_age[,c(1,8,11,13,14,15,16,18,20,21)]

# create a Med_Age variable

data_age <- data_age %>%
   mutate(Med_Age = case_when(
     SUBGROUP_ORDER == 14 ~ 27,
     SUBGROUP_ORDER == 22 ~ 27,
     SUBGROUP_ORDER == 15 ~ 40,
     SUBGROUP_ORDER == 23 ~ 40,
     SUBGROUP_ORDER == 17 ~ 50,
     SUBGROUP_ORDER == 25 ~ 50,
     SUBGROUP_ORDER == 18 ~ 60,
     SUBGROUP_ORDER == 26 ~ 60,
     SUBGROUP_ORDER == 19 ~ 70,
     SUBGROUP_ORDER == 27 ~ 70,
     SUBGROUP_ORDER == 20 ~ 80,
     SUBGROUP_ORDER == 28 ~ 80
   ))


data_age_1990 <- subset(data_age, TIME_PERIOD == "1988-1994")
data_age_2005 <- subset(data_age, TIME_PERIOD == "2005-2008")
data_age_2017 <- subset(data_age, TIME_PERIOD == "2017-3/2020")


data_poverty <- data_poverty[,c(1,8,11,13,14,15,16,18,20,21)]
data_poverty <- subset(data_poverty, !(SUBGROUP_ORDER %in% c(13,16,21,24)) )

# create a numerical variable "Year" to describe average Year within each data collection range

data_poverty <- data_poverty %>%
   mutate(Year = case_when(
      TIME_PERIOD == "1988-1994" ~ 1988,
      TIME_PERIOD == "1995-1998" ~ 1995,
      TIME_PERIOD == "1999-2000" ~ 1999,
      TIME_PERIOD == "2001-2004" ~ 2001,
      TIME_PERIOD == "2005-2008" ~ 2005,
      TIME_PERIOD == "2009-2012" ~ 2009,
      TIME_PERIOD == "2013-2016" ~ 2013,
```

```
      TIME_PERIOD == "2017-3/2020" ~ 2017
   ))
```

1.2.2 Data Table

```
# table for the data

vars <- c("NESTING_LABEL", "Med_Age", "ESTIMATE", "STANDARD_ERROR")

# Function to convert R classes to human-readable types
get_var_type <- function(x) {
  if (is.numeric(x)) {
    "Continuous"
  } else if (is.factor(x) || is.character(x)) {
    "Categorical"
  } else {
    "Other"
  }
}

# Create summary table
variable_summary <- tibble(
  Variable = vars,
  Type = sapply(data_age[vars], get_var_type),
  Class = sapply(data_age[vars], class)
)

# View table
variable_summary
```

```
## # A tibble: 4 x 3
##   Variable       Type        Class
##   <chr>          <chr>       <chr>
## 1 NESTING_LABEL  Categorical character
## 2 Med_Age        Continuous  numeric
## 3 ESTIMATE       Continuous  numeric
## 4 STANDARD_ERROR Continuous  numeric
```

1.2.3 Data Visualization

```
# plot

data_age_1990$Year <- "1990"
data_age_2005$Year <- "2005"
data_age_2017$Year <- "2017"

combined <- rbind(data_age_1990, data_age_2005, data_age_2017)
combined <- combined %>%
  mutate(group = interaction(Year, NESTING_LABEL))
```

```
BP_AGE_GENDER <- ggplot(combined,
       aes(x = Med_Age,
           y = ESTIMATE,
           color = group,
           group = group)) +
  geom_point(size = 3, alpha = 0.6) +
  geom_line() +
  geom_errorbar(aes(
    ymin = ESTIMATE - 1.96 * STANDARD_ERROR,
    ymax = ESTIMATE + 1.96 * STANDARD_ERROR
  ),
  width = 0.6) +
  labs( title= "Percentage of hypertensive population per age group (bins of 10 yrs)",
  subtitle="(CDC data collected for time frames 1988-94 and 2017-20)",
  x = "median age for 10-year age group",
  y = "Estimate of hypertensive (%)")

# save plot to a output file
ggsave(
  filename = "../output/Hypertension_Age_Gender.png",
  plot = BP_AGE_GENDER,
  width = 8.5,
  height = 10,
  dpi =300
)
```

1.3 Hypertension as a function of Income levels

Does hypertensive disease affect all individuals the same regardless of income?

1.3.1 Data Visualization

```
data_poverty$SUBGROUP <- factor(
  data_poverty$SUBGROUP,
  levels = c("<100% FPL",
             "100% to <200% FPL",
             "200% to <400% FPL",
             " 400% FPL")
)

# bar plot to show hypertensive estimates over time periods of collection.

BP_POV2 <- ggplot(data_poverty,
       aes(x = TIME_PERIOD,
           y = ESTIMATE,
           fill = SUBGROUP)) +
  geom_col(position = "dodge") +
  geom_errorbar(
    aes(
      ymin = ESTIMATE - 1.96 * STANDARD_ERROR,
```

```
      ymax = ESTIMATE + 1.96 * STANDARD_ERROR
    ),
    position = position_dodge(width = 0.9),
    width = 0.2
  ) +
scale_fill_manual(
  values = c("red", "orange", "yellow", "darkgreen"),
  name = "Poverty level (% FPL)",
  labels = c("<100", "100-199", "200-399", " 400")
) +
  labs(
    title = "Evolution of hypertension prevalence by poverty level",
    subtitle = "(CDC data, age-adjusted)",
    x = "Time period",
    y = "Estimate (%) with 95% CI",
    fill = "Poverty level (% FPL)"
  )
```

```
# save plot to a output file
ggsave(
  filename = "../output/Hypertension_Poverty_barplot.png",
  plot = BP_POV2,
  width = 8.5,
  height = 10,
  dpi =300
)
```

2 Conclusions

The prevalence of hypertension between men and women is different between ages 18 and 59(ref 1). However, the rate at which women develop hypertension starting age 18 is much steeper compared to men (ref 2). A disparity in the prevalence of hypertension is also observed for the four income level analyzed: this trend seems to indicate that hypertension is more prevalent for individuals at poverty levels up to 399% of the Federal Poverty Level (ref 3).

3 References

- 1. Connelly PJ, Currie G, Delles C. Sex Differences in the Prevalence, Outcomes and Management of Hypertension. Curr Hypertens Rep. 2022 Jun,24(6):185-92. doi: 10.1007/s11906-022-01183-8
- 2. Ostchega Y, Fryar CD, Nwankwo T, Nguyen DT. Hypertension prevalence among adults aged 18 and over: United States, 2017–2018. NCHS Data Brief. 2020;364:1–8.
- 3. Hardy ST, Jaeger BC, Foti K, Ghazi L, Wozniak G, Muntner P. Trends in Blood Pressure Control among US Adults With Hypertension, 2013–2014 to 2021–2023. Am J Hypertens. 2025;38(2):120-128. doi:10.1093/ajh/hpae141